

Quiz 3 Section 1 Grading

Context: A factory produces small electronic components, each with probability $p = 0.04$ of being defective.

Tasks and Full Answers

Task 1 — Distribution (5 pts)

Question: Write down the probability distribution for the number X of defective components in n independent trials.

Answer:

The number of defective components follows a **binomial distribution**:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \dots, n.$$

Task 2 — Exactly 3 defectives in 20 (5 pts)

Question: Find $P(X = 3)$ for $n = 20$, $p = 0.04$.

Step-by-step:

1. Use the binomial PMF:

$$P(X = 3) = \binom{20}{3} (0.04)^3 (0.96)^{17}.$$

2. Compute components:

- $\binom{20}{3} = 1140$
- $(0.04)^3 = 0.000064$
- $0.96^{17} \approx 0.5133$

3. Multiply:

$$P(X = 3) \approx 1140 \times 0.000064 \times 0.5133 \approx 0.0375$$

Final Answer:

$P(X = 3) \approx 0.0375 \text{ (3.75\%)}$

Task 3 — At most 2 defectives in 20 (5 pts)

Question: Find $P(X \leq 2)$.

Step-by-step:

1. $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$

2. Compute individually:

- $P(X = 0) = 0.96^{20} \approx 0.4420$
- $P(X = 1) = \binom{20}{1} (0.04)(0.96^{19}) \approx 20 \cdot 0.04 \cdot 0.4604 \approx 0.3683$
- $P(X = 2) = \binom{20}{2} (0.04)^2 (0.96^{18}) \approx 190 \cdot 0.0016 \cdot 0.4792 \approx 0.1457$

3. Sum:

$$P(X \leq 2) \approx 0.4420 + 0.3683 + 0.1457 = 0.956$$

Final Answer:

$$P(X \leq 2) \approx 0.956 \text{ (95.6\%)}$$

Task 4 — Poisson Approximation for n=200, p=0.04 (5 pts)

Step 1: Compute expected number of defectives:

$$\lambda = n \cdot p = 200 \cdot 0.04 = 8$$

Step 2: Poisson formula for exactly 4 defectives:

$$P(X = 4) \approx \frac{e^{-\lambda} \lambda^4}{4!} = \frac{e^{-8} \cdot 8^4}{24}$$

Step 3: Compute numerically:

- $8^4 = 4096$
- $4! = 24$
- $4096/24 \approx 170.6667$
- $e^{-8} \approx 0.0003355$
- Multiply: $0.0003355 \cdot 170.6667 \approx 0.0573$

Final Answer:

$$P(X = 4) \approx 0.0573 \text{ (5.73\%)}$$

Grading Rubric (Total 20 pts)

Task 1 — 5 pts

- **2 pts:** Correctly identify the distribution as binomial and define X .
- **2 pts:** Correct PMF formula $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$.
- **1 pt:** Correct domain $k = 0, 1, \dots, n$.

Partial credit: 1–3 pts for correct identification or formula.

Task 2 — 5 pts

- **2 pts:** Correct formula substitution with $n = 20$, $k = 3$, $p = 0.04$.
- **1 pt:** Correct calculation of $\binom{20}{3} = 1140$.
- **2 pts:** Correct numeric result (within ± 0.001 of 0.0375).

Partial credit: 2–4 pts for formula correct but arithmetic errors.

Task 3 — 5 pts

- **1 pt:** Recognize $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$.
- **2 pts:** Correct computation of individual terms.
- **2 pts:** Correct sum within ± 0.001 of 0.956.

Partial credit: 2–3 pts for partial sum or missing one term.

Task 4 — 5 pts

- **1 pt:** Correctly compute $\lambda = n \cdot p = 8$.
- **2 pts:** Correct Poisson formula: $P(X = k) = e^{-\lambda} \lambda^k / k!$.
- **2 pts:** Correct numeric evaluation within ± 0.001 of 0.0573.

Partial credit: 2–3 pts if formula correct but calculation slightly off; 1 pt if approximate reasoning given.

End of answer key and grading guideline.